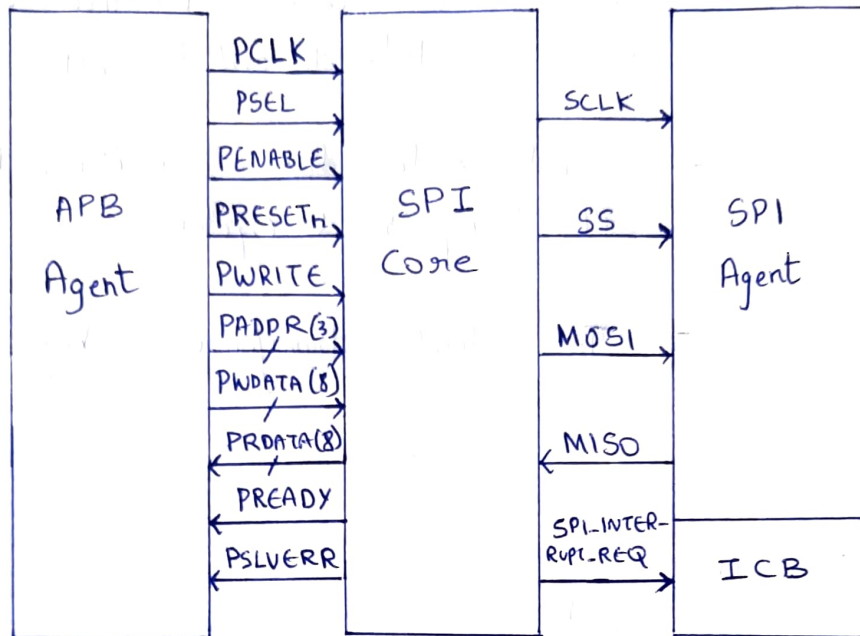


APB Based SPI Verification

V-plan

Block diagram:Signals Involved:

PCLK - Clock signal. All APB signals are timed against the rising edge of PCLK.

PRESETn - Active-low reset signal

PADDR - 3-bit signal for addressing SPI core register

PSEL - Signal to select the SPI core

PENABLE - Indicates the second and subsequent transfer

PWRITE - Indicate Write Access when High and Read when Low.

PWDATA - 8-bit wide data bus, driven by APB during write cycles

PREADY - Indicator SPI core's Readiness for transfer of Data to and from APB

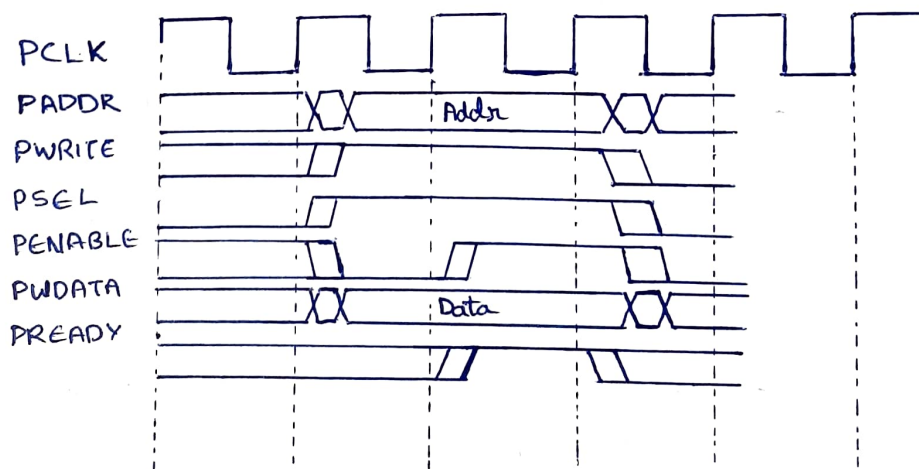
PRDATA - 8-bit wide data bus, driven by SPI during read cycles

PSLVERR - To indicate transfer error by SPI

### Features of SPI

- 1) Resettable
- 2) Only Master mode
- 3) Full Duplex Serial Communication of 8-bit Data
- 4) Configurable Clock Frequency
- 5) Programmable Serial Clock Polarity and phase.
- 6) 4 modes of Operation for Data transfer considering LSB and MSB
- 7) Low Power mode

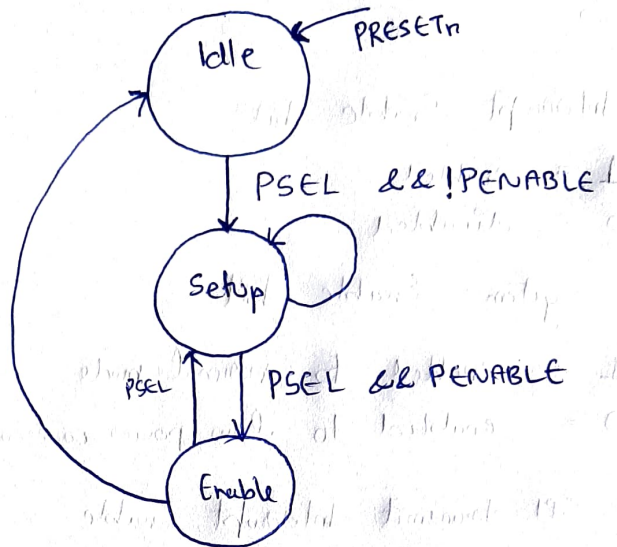
### Waveform



WRITE transfer  
with no wait  
states.

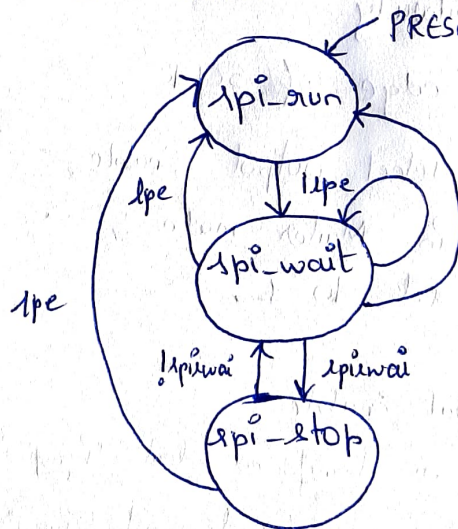
## APB Phases

- 1) Idle - Indicates APB not selecting any peripheral
- 2) Setup - Indicates APB selected a Core.
- 3) Enable - Indicates transfer to the selected core



## SPI modes

- 1) Run - Indicates normal Operation
- 2) Wait - Indicates Low power mode
- 3) Stop - Indicate SPI system disabled for Communication with slave



## Register Frame format

CR1, Address = 3'b000, R/W

Reset:

| SPIE | SPE | SPTIE | MSTR | CPOL | CPHA | SSOE | LSBFE |
|------|-----|-------|------|------|------|------|-------|
| 0    | 0   | 0     | 0    | 0    | 1    | 0    | 0     |

SPIE - Interrupt Enable bit

1 = enabled

0 = disabled

SPE - System Enable bit

1 = enabled to normal mode

0 = enabled to low power conservative mode

SPTIE - SPI transmit Interrupt Enable

1 - SPTIEF interrupt enabled

0 - SPTIEF interrupt disabled

MSTR - SPI Master / Slave Mode bit

1 - Master mode

0 - Slave mode

CPOL - SCLK polarity

CPHA - Active edge of SCLK

SSOE - Slave select output Enable

Enabled only in Master mode, SS is the core

Output if set to 1.

LSBFE - LSB-first Enable

1 - Data is transferred least significant bit first.

0 - Data is transferred most significant bit first.



CR2, Address = 3'b001, RW

|        |   |   |   |        |        |   |         |      |
|--------|---|---|---|--------|--------|---|---------|------|
|        | 0 | 0 | 0 | MODFEN | BIDROE | 0 | SPISWAI | SPCO |
| Reset: | 0 | 0 | 0 | 0      | 0      | 0 | 0       | 0    |

MODFEN - Mode Fault Enable bit

- 1 → SS port pin with MODF feature
- 0 → SS port pin is not used by the SPI

BIDROE - If SPCO is low, then this bit decides whether to transmit or receive.

- 1 → output buffer enabled
- 0 → output buffer disabled

SPISWAI - SPI stop in wait Mode bit

- 1 → Stops SCLK
- 0 → SCLK operates normally.

SPCO - Serial Pin Control Bit 0

Enable bidirectional pin configuration

- if 0 → full duplex
- 1 → Half duplex

Baud Rate Register Address = 3'b010, RW

|       |   |      |      |      |   |      |      |      |
|-------|---|------|------|------|---|------|------|------|
|       | 0 | SPR2 | SPR1 | SPR0 | 0 | SPR2 | SPR1 | SPR0 |
| Reset | 0 | 0    | 0    | 0    | 0 | 0    | 0    | 0    |

SPR2 - SPR0 ⇒ SPI Baud Rate Biselection bits

SPR2 - SPR0 ⇒ SPI Baud Rate Selection bits

$$\text{Baud Rate Divisor} = (SPR + 1)_2 \cdot (SPR + 1)$$

Band Rate = Bus Clock / Band Rate Divisor!!

SPI Status Register Address = 3'b011 , R

|       |      |   |        |       |   |   |   |   |
|-------|------|---|--------|-------|---|---|---|---|
|       | SPIF | 0 | SPTIEF | MODEF | 0 | 0 | 0 | 0 |
| Reset | 0    | 0 | 0      | 0     | 0 | 0 | 0 | 0 |

SPIF - SPI Interrupt flag

1 - New data copied to SPIDR

0 - Transfer not yet complete

SPTIEF - SPI Transmit Empty Interrupt Flag

1 - SPI DR empty

0 - SPI DR not empty

MODEF - Mode Fault Flag

1 - Mode fault has occurred

0 - Mode fault has not occurred

SPI Data Register Address = 3'b101 , R/W

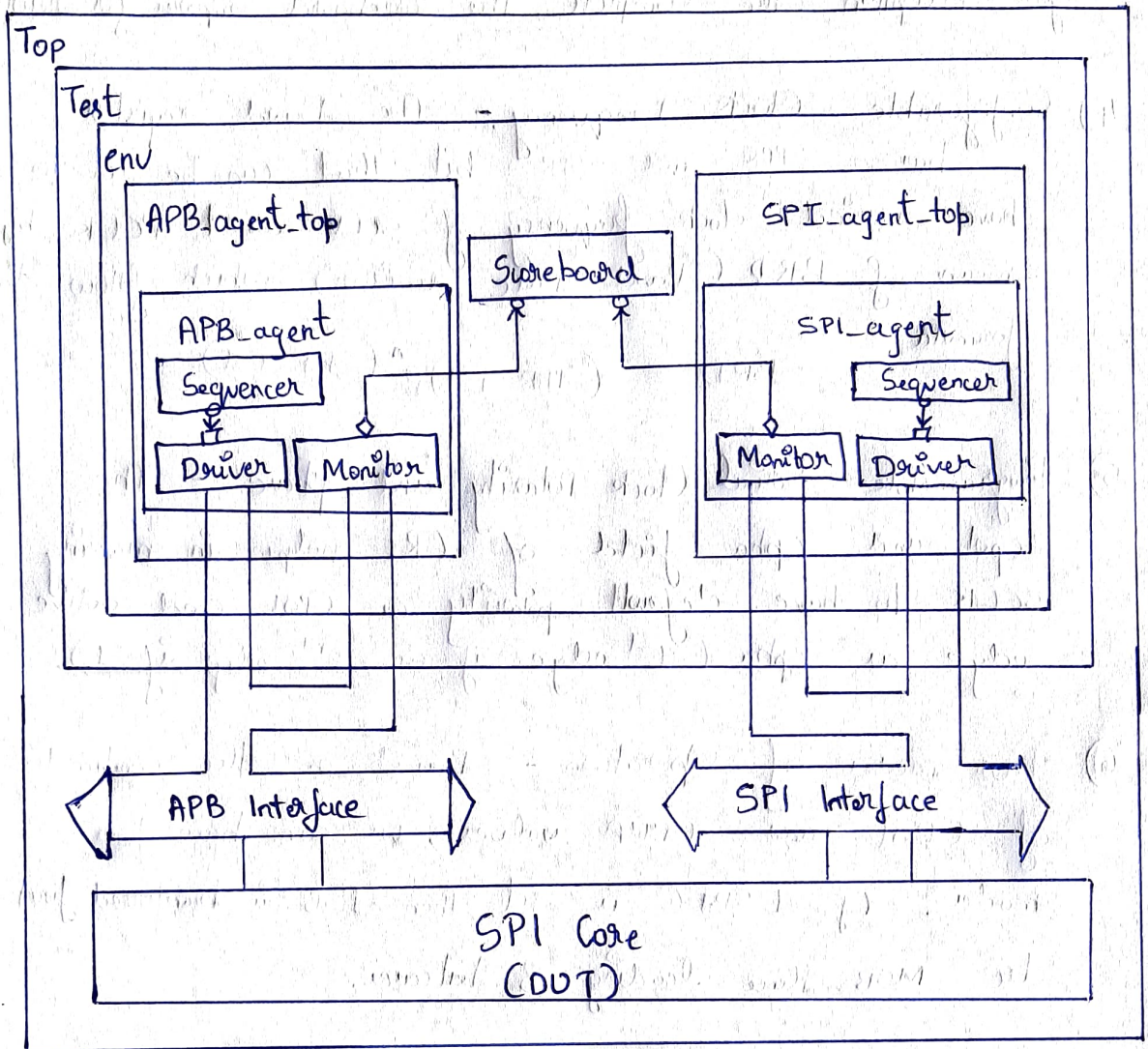
|       |       |   |   |   |   |   |   |   |
|-------|-------|---|---|---|---|---|---|---|
|       | Bit 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 |
| Reset | 0     | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

Read : anytime ; normally read only after SPIF is set

Write : anytime.



## TB Architecture



## Implementation

- 1) **Resetability** - **PRESETn** is brought low for asynchronous reset. The SPI Core registers should be reset.
- 2) **Master mode** - The **master** field of **CRI** is always set to 1.
- 3) **Full Duplex** - While the slave is selected (**ss=1'b0**), then **tx** is undertaken. The SPI Agent

enable to sense it and in turn provides miso data for Reception. Once Complete, the scoreboard compares for result.

4) Configurable Clock Frequency.- The Baud register has SPPR and SPR bit that can be set to have varied clock frequency of SCLK with PCLK by means of BRD (Baud Rate divider) which follows the formula

$$BRD = (SPPR + 1)2^n (SPR + 1)$$

5) Programmable Serial Clock Polarity and phase - The cpol and cpha field of CR1 helps in making SCLK to have default polarity as CPOL and active edge as cpha (1<sup>st</sup> edge if 0, 2<sup>nd</sup> edge if 1).

6) Four modes of Operation - Based on the combination of EPOL and CPHA values, we can have four modes. If LSBFE is set, then LSB is transferred first else MSB. Hence leading 8 bit transfer.

7) Low Power mode - If txn is in progress and spe field of CR1 is low, then the SPI is said to operate in SPI wait mode. Thus, if spiswai field of CR2 is asserted, then the core enters stop mode and the transmission/reception terminates. SCLK is stopped generating.



## Test Case

### Positive test Case

SPI in normal mode of Operation, performing test on combination of CPOL, CPHA & LSBFE. Finally comparing in Scoreboard.

| SL.No. | LSBFE | CPOL | CPHA |
|--------|-------|------|------|
| 1      | 0     | 0    | 0    |
| 2      | 0     | 0    | 1    |
| 3      | 0     | 1    | 0    |
| 4      | 0     | 1    | 1    |
| 5      | 1     | 0    | 0    |
| 6      | 1     | 0    | 1    |
| 7      | 1     | 1    | 0    |
| 8      | 1     | 1    | 1    |

### Negative test Case

Low Power mode, SPI in wait mode by having 1st bit low in CRI.

Any transmission or Reception in progress will be aborted if spiwait bit of CR2 is asserted.

### Corner test Case

Adjusting the Baud Rate Divisor value by Randomizing SPPR & SPR.

Performing Simultaneous two Writes onto the DR, terminate transmission.

## Covergroup / Functional Coverage

For APB

PADDR = {0, 1, 2, 3, 5}

PRESETn = {0, 1}

PWRITE = {0, 1}

PWDATA = { binx bl = { [8'h00 : 8'd12] }  
                  binx bl = { [8'd128 : 8'hff] } }

PRDATA = PWDATA

PREADY = {1}

PSLVERR = {1, 0}

Cover PWRITE, PADDR, PWDATA

For SPI

SS = {0, 1}      SELK = {0, 1}

MOSI = {0, 1}

MISO = {1, 0}

## Assertion

- 1) @ (posedge PCLK) (PSEL && !PENABLE)  $\rightarrow$  ## [1:2] PENABLE
- 2) @ (posedge PCLK) (PSEL && !PENABLE)  $\rightarrow$  PSEL until PENABLE
- 3) @ (posedge PCLK) PENABLE  $\rightarrow$  ## [1:2] PREADY
- 4) @ (posedge PCLK) PREADY  $\Rightarrow$  (!PENABLE && !PREADY)

1  $\rightarrow$  Penable, 2  $\rightarrow$  PSEL, 3  $\rightarrow$  Pready, 4  $\rightarrow$  Pready  
and Penable went low together

Penable should be high within 1 to 2 clock cycles PSEL is high

PSEL should remain high until Penable is asserted

Once PENABLE High, within two cycles PREADY should be asserted

Once PREADY High, PENABLE & PREADY should go low together.